



Comparing Low Dosages of ABA Treatment on Children's Treatment Gains and School Readiness

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Abstract

Objectives The current study analyzed the data from a 1-year pragmatic clinical trial on the effectiveness of a naturalistic implementation of the PEAK Relational Training System in a public school setting.

Methods We accessed student records from two classrooms at the same grade level within one public school. Students' assignments in these two classrooms were based on the severity of their diagnosis. Students in both classrooms received conventional special education services and applied behavior analysis intervention based on the PEAK Relational Training System. Post hoc analyses were conducted after one school year to compare students' behavior skill levels, school readiness skills, and their relationship with their daily dosage of ABA intervention.

Results Post hoc analyses indicated a significant difference in students' behavior skill level and school readiness skills at the beginning of the school year. Significant differences were also found in the amount of PEAK interventions delivered to each classroom. At the end of the school year, all students demonstrated improvements on the PEAK pre-assessment ($p < .001$) and the Bracken School Readiness Assessment (BSRA; $p < .001$). A secondary analysis examined variables that predicted students' progress and showed that their daily ABA dosage ($R^2 = .278$) was a significant predictor of their BSRA improvement.

Conclusions The current study added to a growing body of research demonstrating the utility and feasibility of implementing PEAK-based interventions in special education settings. Behavior interventions along with special education practices produced significant outcomes on students' overall skill level and school readiness. Implications of applying a low dosage of ABA intervention are discussed.

Keywords PEAK Relational Training System · Bracken School Readiness Assessment · Special Education · Autism Treatment · ABA Dosage

Teachers in school settings often face scheduling challenges when working with students with autism and other intellectual and developmental disabilities. In order to address the total needs of the student, a finite amount of learning time during a school day must be carefully distributed across many areas beyond core academic content (e.g., English Language Arts, Mathematics). These areas may include physical education, lunch and recess, speech language, occupational therapy, physical therapy, social work, and potentially applied behavior

analysis (ABA). The combination of such services is obviously related to the needs of the individual student, thus leaving the ideal often unattainable and the practical in question. The uniqueness of each case requires teachers to individually assess their students and develop tailored treatment plans that account for students' strengths and weaknesses (Keen et al., 2016). Yet when a need for ABA may be justified, impressions on the practicality of implementation may be overwhelming. Researchers have reported that teachers face a wide range of challenges when instructing this population, such as the lack of training and resources, barriers to creating an inclusive environment, and the inconsistency between school-based and community/clinic-based interventions (Hedges et al., 2014; Lindsay et al., 2013).

Despite that a variety of evidence-based Comprehensive Treatment Models (CTMs) have been developed and verified within the field of ABA, very few attempts have been made

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